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RiskNova – AI-Driven Project Risk Assessment

RiskNova is an AI-powered web application that estimates the risk level of software projects using a trained Deep Neural Network (DNN). The model is a supervised machine learning classifier that learns from structured project attributes such as complexity, team size, communication, cost, testing quality, and previous defects to classify the project as Low, Medium, or High risk.

Accuracy

Accuracy 95.29%

Built With

Built With Django 6.1, TensorFlow, Keras, scikit-learn, NumPy, Joblib

License

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Project Highlights

-  Predicts project risk from structured project metadata
-  Uses a trained 15-feature DNN (Deep Neural Network) for multiclass risk classification
-  Provides risk probability, confidence score, and mitigation recommendations
-  Saves historical predictions in SQLite for dashboard tracking
-  Deployed as a Django web application with a responsive UI
-  Includes a dashboard for recent predictions and risk distribution
-  Supports evaluation of model performance and decision support for project managers

Demo

RiskNova evaluates a project's risk using a trained AI model and returns an actionable result with contributing factors and mitigation strategies.

You can test it using the Django-based web app.

Screenshot

Upload project information and click predict to receive the risk result.

- → Get predicted risk level
- → Get probability and confidence
- → View top contributing factors
- → Receive recommendations

Video Demo

 [\[Click here to view the demo\]](#)

► Watch Demo

Dataset

Size: 15 engineered project attributes for risk classification

Labels: Low, Medium, High risk

Features include:

- Project size
- Team size
- Project duration
- Estimated cost
- Project complexity
- Requirements stability
- Requirement changes
- Communication level
- Project management quality
- Resource availability
- Team experience
- Technical expertise
- Testing level
- Quality factors
- Previous defects

The model is trained on historical project data and saved as a reusable artifact for prediction.

Tools & Technologies

Category	Tools / Frameworks
Programming	Python
Web Framework	Django 6.1
Deep Learning	TensorFlow, Keras
Data Processing	NumPy, Pandas
ML Utilities	scikit-learn, Joblib
Database	SQLite
Frontend	HTML, CSS, Bootstrap
IDE	VS Code, Jupyter Notebook

Performance Summary

Model	Risk Accuracy	Notes
DNN Project Risk Model	95.29%	Deep neural network trained for 15-feature project risk classification
Baseline Logistic Model	Lower	Simple traditional baseline
Rule-based scoring	Lower	Less robust for nonlinear patterns

 Risk Mapping

Risk Level	Meaning
LOW	Healthy project with manageable risk
MEDIUM	Needs active monitoring and mitigation
HIGH	Requires immediate planning and risk control

 Evaluation Metrics

- Accuracy: 95.29%
- Model Type: DNN (Deep Neural Network), a machine learning model based on neural networks
- Output: Probability scores for each class (LOW, MEDIUM, HIGH) with confidence value
- Inference Speed: Fast enough for web-based use after warm-up
- Output includes important factors and recommendations

 Web Interface Pages

-  Home: Introduction and project overview
-  Predict: Fill in project data and generate risk score
-  Dashboard: View historical predictions and distribution
-  Result: Show risk level, probabilities, factors, and recommendations

 Installation Guide

```
git clone <your-repository-url>
cd New folder
```

Create virtual environment

```
py -3.13 -m venv .venv
.\.venv\Scripts\Activate.ps1
```

Install dependencies

```
pip install django tensorflow scikit-learn joblib numpy
```

Run the application

```
python manage.py migrate  
python manage.py runserver
```

Open the browser at:

```
http://127.0.0.1:8000/
```

🌟 Future Work

- Expand dataset with more real-world project records
- Add explainable AI for each prediction factor
- Deploy as a cloud-hosted production app
- Add project team recommendations and risk alerts
- Improve forecasting for large enterprise project portfolios

👥 Authors

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📄 License

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🌟 Acknowledgements

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- Team members who contributed to model validation and testing